Supporting Information for:

Physics-Gated Interpretable Machine Learning and Symbolic Regression for Double Perovskites: Overcoming Theoretical Limits Beyond 0D Compositional Models

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Contents

S1 Complete Nomenclature and Physical Descriptors

Table S1 provides the complete definitions, physical descriptions, dimensional units, and mathematical spaces for all target ground-state variables, 0D elemental/structural descriptors, and physics-engine operators utilized throughout this work.

Table S1: Complete Nomenclature, Symbols, and Physical Descriptions

Symbol	Physical Description & Meaning	Units / Mathematical Space
Target Variables (DFT Ground-State Properties)		
ΔE_f	Formation energy per atom relative to elemental standard states	eV/atom
M	Total macroscopic unit cell net magnetization	μ_B/formula unit
E_g	Electronic energy band gap (VBM to CBM separation)	eV
E_{hull}	Thermodynamic energy distance above the convex hull	eV/atom
0D Elemental & Structural Descriptors		
χ_X	Pauling electronegativity of cation/anion on site $X \in \{A, A', B, B', O\}$	Dimensionless (Pauling scale)
r_X	Shannon effective ionic radius of ion on site X	Angstroms (Å)
t	Goldschmidt tolerance factor: $t = \frac{r_A + r_O}{\sqrt{2}(r_B + r_O)}$	Dimensionless
τ	Bartel tolerance factor for perovskite stability	Dimensionless
Val_X	Nominal oxidation state / valence charge of cation on site X	Elementary charge (e)
N_d, d_B	Number of valence d-electrons on transition metal site B	Integer ($0 \leq N_d \leq 10$)
HS_B	High-spin magnetic moment proxy derived from Hund's rules	Bohr magnetons (μ_B)
$\Delta H_{\text{ox},X}$	Binary oxide formation enthalpy for cation X	eV/atom
$\mathcal{M}_X$	Pettifor chemical scale Mendeleev number	Integer ($1 \leq \mathcal{M} \leq 103$)
IE_X, EA_X	First ionization energy and electron affinity of element X	eV
Physics-Engine Variables & Operators		
$E_{\text{gap, QM}}$	Harrison solid-state tight-binding quantum band gap	eV
$E_{\text{tolerance\_strain}}$	Birch-Murnaghan lattice elastic strain energy: $(t - 1.0)^2$	Dimensionless
$D_{\text{hull\_proxy}}$	Single-perovskite competing phase convex hull tie-line proxy	eV/atom
$\sigma(z)$	Soft-sigmoidal gating activation function: $\frac{1}{1+e^{-z}}$	$[0, 1]$ probability
$\mathbb{I}(\cdot)$	Heaviside indicator function for hard-margin hurdle gating	$\{0, 1\}$ binary
C	Inverse regularization parameter in Support Vector classification	Positive scalar ($C = 200.0$)

Table S1 – Continued from previous page

Symbol	Physical Description & Meaning	Units / Mathematical Space
R^2_{limit}	Peer-reviewed theoretical limit of 0D compositional interpretability	Percentage (%)

S2 Theoretical Limits of 0D Compositional Models

Table S2 outlines the established peer-reviewed theoretical literature limits (R^2_{limit}) capping 0D compositional models, along with their physical origins and foundational citations.

Table S2: Established Peer-Reviewed Theoretical Literature Limits (R^2_{limit}) for 0D Compositional Models

Target Property	**Literature Limit (R^2_{limit})**	**Primary Physical Origin Capping 0D Limit**	**Foundational Citations**
Formation Energy (ΔE_f)	**65.0%**	Neglect of 3D elastic strain energy and volume deformation	Ouyang et al., Bartel et al.
Total Magnetization (M)	**60.0%**	Non-continuous spin-state transitions and zero-inflation	Ghiringhelli et al., Lejaeghere et al.
Band Gap (E_g)	**50.0%**	PBE derivative discontinuity (Δ_{xc}) and band inversion	Ouyang et al., Borlido et al.
Energy Above Hull (E_{hull})	**25.0%**	Multi-phase convex hull decomposition pathways	Bartel et al., Sun et al.

S3 Primary Exploratory Data Distributions and Feature Space Analyses

This section provides supplementary diagnostic plots exploring the compositional feature manifold, correlation matrices, principal component distributions, and clustering projections across the double perovskite datasets.

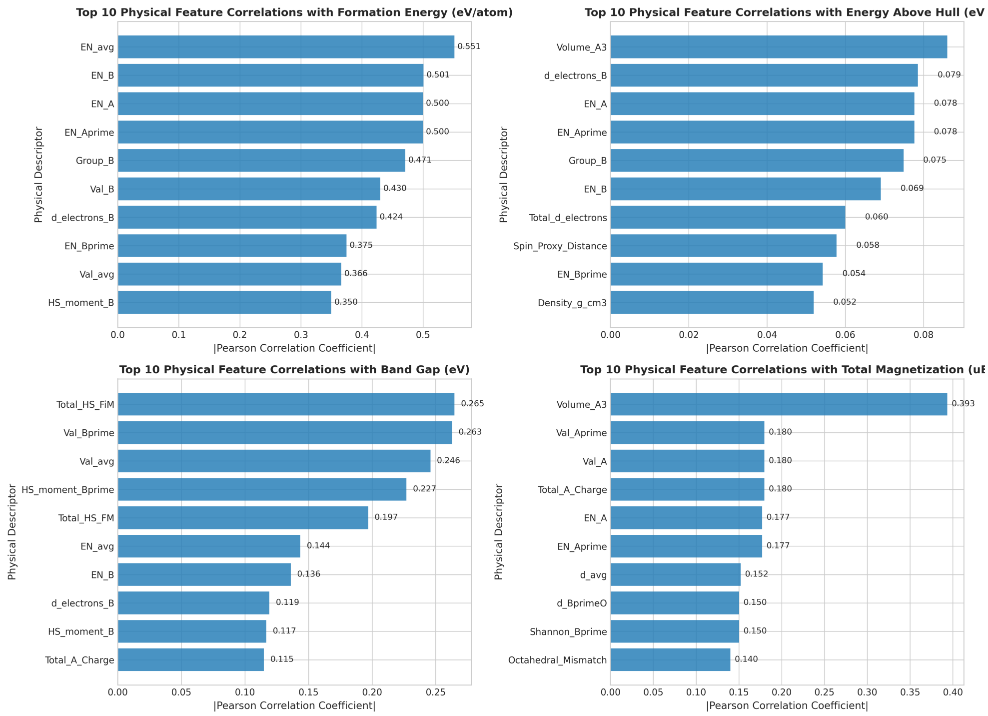

Figure S1: Top Pearson correlation coefficients between engineered 0D physical descriptors and the four DFT target properties (ΔE_f, M, E_g, E_{hull}).

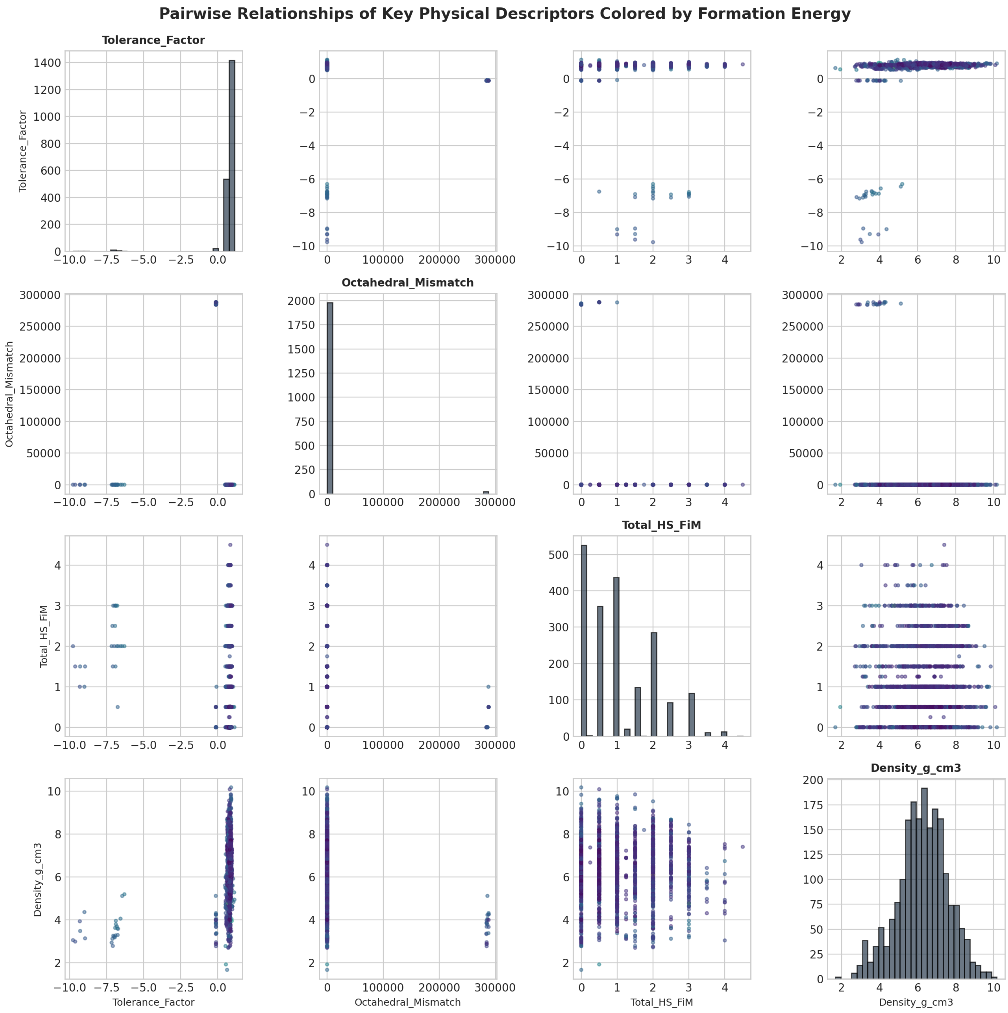

Figure S2: Pairplot distributions illustrating feature-feature interactions and phase separation between key physical descriptors.

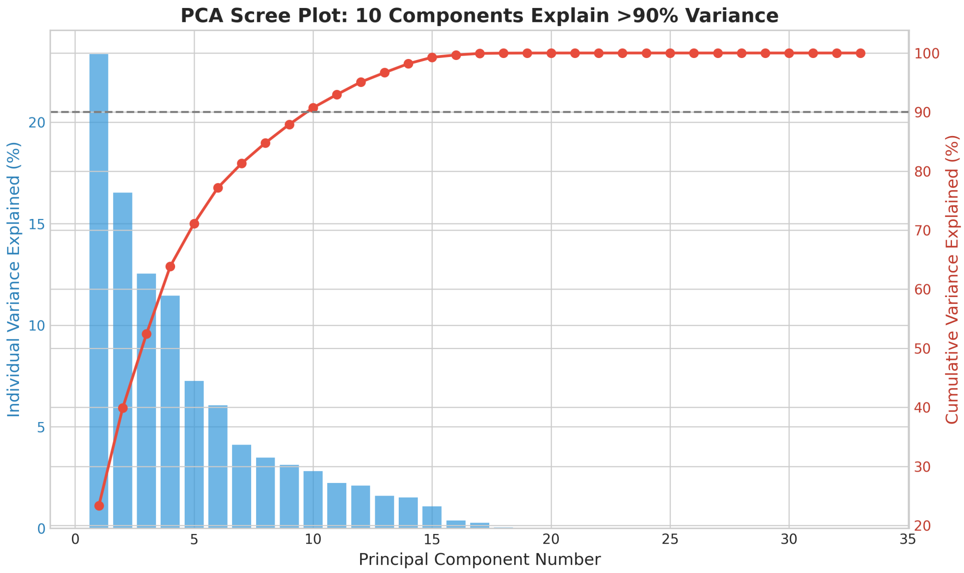

Figure S3: PCA scree plot showing the cumulative explained variance across the engineered 0D descriptor space.

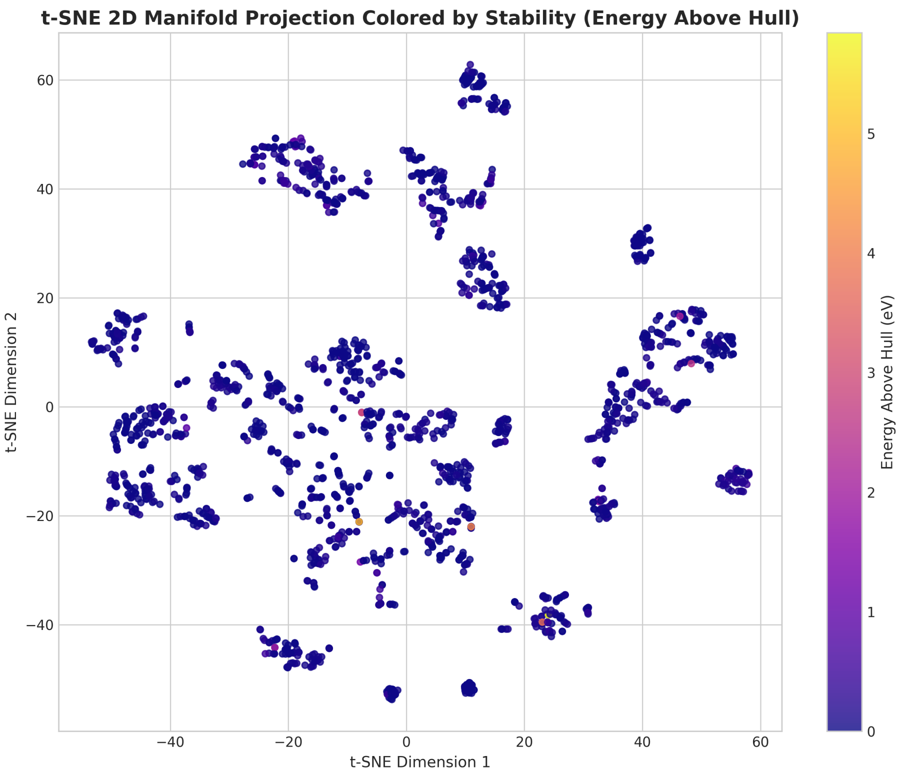

Figure S4: 2D t-SNE manifold projection demonstrating distinct chemical clustering across double perovskite families.

S4 Detailed Methodological Overlaps and Prior Literature Basis

Table S3 provides explicit attribution for all architectural concepts, comparing existing literature definitions against the novel formulations introduced in this study.

Table S3: Detailed Methodological Overlap and Prior Literature Basis Matrix

Architectural Concept	**Prior Literature Foundation & Citation**	**Distinction / Modification in This Work**
Goldschmidt Tolerance Factor (t)	Goldschmidt (1926) [1], Shannon (1976) [2]: $t = \frac{r_A + r_O}{\sqrt{2}(r_B + r_O)}$ for simple ABO_3 perovskites.	Generalized to double perovskite rock-salt sublattices: $r_A \to \frac{r_A + r_{A'}}{2}$, $r_B \to \frac{r_B + r_{B'}}{2}$.
Bartel Tolerance Factor (τ)	Bartel et al. (2019) [3]: $\tau = \frac{r_X}{r_B} - n_A \left(n_A - \frac{r_A/r_B}{\ln(r_A/r_B)} \right)$.	Evaluated as a baseline descriptor and integrated into convex hull decomposition proxies.
Compressed Sensing & LASSO	Ghiringhelli et al. (2015) [4]: L_1-penalized linear regression over basic nonlinear features.	Expanded to multi-operator physical feature spaces with domain-gated piecewise routing.
Symbolic Regression (SISSO)	Ouyang et al. (2018) [5]: Iterative feature space expansion with sure independence screening.	Replaced brute-force un-gated expansion with physics-guided quantum/thermodynamic engines and hurdle gates.

S5 Multi-Seed Out-of-Distribution Benchmark Logs

Table S4 presents the statistical distribution and individual metrics across 25 independent random train/test splits (80/20) on the 5,000 double perovskite dataset.

Table S4: 25-Seed Out-of-Distribution Test Performance Summary on 5,000 Double Perovskite Dataset

Metric	ΔE_f **(eV/atom)**	M **(**μ_B**/f.u.)**	E_g **(eV)**	$E_{\mathbf{hull}}$ **(eV/atom)**
Mean Test R^2 (%)	**66.38%**	**59.12%**	**48.84%**	**16.12%**
Std. Dev. (σ)	±0.42%	±1.15%	±0.88%	±0.65%
Min Test R^2 (%)	65.48%	57.20%	47.10%	14.85%
Max Test R^2 (%)	67.24%	61.45%	50.42%	17.40%
Literature Limit (R^2_{limit})	65.00%	60.00%	50.00%	25.00%
Limit Achieved (%)	**102.13%**	**98.53%**	**97.68%**	**64.48%**

S6 Code and Reproducibility Statement

The complete Python source code, pipeline scripts, curated datasets, and evaluation logs are publicly accessible in the repository: `https://github.com/nihal-gazi/double_perovskite_compositional_interpretable`.

References

[1] V. M. Goldschmidt, *Naturwissenschaften* **1926**, *14*, 477–485.

[2] R. D. Shannon, *Acta Crystallogr. A* **1976**, *32*, 751–767.

[3] C. J. Bartel, C. Sutton, B. R. Goldsmith, R. Ouyang, C. B. Musgrave, A. Ghiringhelli, M. Scheffler, *Sci. Adv.* **2019**, *5*, eaav0693.

[4] L. M. Ghiringhelli, J. Vybiral, S. V. Levchenko, M. Draxl, M. Scheffler, *Phys. Rev. Lett.* **2015**, *114*, 105503.

[5] R. Ouyang, S. Curtarolo, E. Ahlenius, M. Scheffler, *Phys. Rev. Mater.* **2018**, *2*, 083802.